

- Q.29 What is convection, explain the type of convection.
 Q.30 Write about Seider and Tate's equation.
 Q.31 Explain Buckingham Pi theorem and its application in heat transfer.
 Q.32 Write down about fouling factor.
 Q.33 Explain thermal conductivity and thermal conductance with units and physical significance.
 Q.34 Explain the concept of exchange of energy between two parallel planes of different emissivity.
 Q.35 Derive the Stefan-Boltzmann law and explain black body radiation.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe in detail the construction and working of shell and tube heat exchanger with labeled diagram.
 Q.37 Derive the expression for critical thickness of insulation for a cylinder and explain its physical significance.
 Q.38 Discuss problems in evaporators such as:
- Entrainment
 - Foaming
 - Scaling

No. of Printed Pages : 4

180643/120643/116843

Roll No.

**4th Sem / P&P, Chem Engg.
 (spl. Paint Tech.), Chem Engg. (Spl. Polymer Engg.)
 Sub : Heat Transfer**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Fourier's law of heat conduction states that heat transfer rate is proportional to:
- a) Temperature
 - b) Temperature gradient
 - c) Thermal conductivity only
 - d) Area only
- Q.2 Unsteady state conduction occurs when:
- a) Temperature does not change with time
 - b) Heat flow is zero
 - c) Temperature changes with time
 - d) Thermal conductivity is constant
- Q.3 Grashof number is important in :
- a) Forced convection
 - b) Radiation
 - c) Natural convection
 - d) Conduction
- Q.4 Entrainment in evaporators refers to:
- a) Liquid droplets carried with vapor
 - b) Steam condensation
 - c) Scale formation
 - d) Foam destruction

- Q.5 Finned tube heat exchangers are used when:
- Heat transfer coefficient is low on one side
 - Both fluids are liquids
 - No temperature difference exists
 - Radiation dominates
- Q.6 In backward feed multiple effect evaporator:
- Feed enters first effect
 - Feed enters last effect
 - Steam enters last effect
 - No feed enters
- Q.7 The purpose of insulation is to:
- Increase heat loss
 - Prevent heat loss or gain
 - Increase temperature
 - Increase radiation
- Q.8 Heat loss from an insulated pipe initially:
- Always decreases
 - Always increases
 - First increases then decreases
 - Remains constant
- Q.9 Scale formation in heat exchanger result in:
- Increased heat transfer
 - Reduced thermal resistance
 - Decreased heat transfer
 - Increased velocity
- Q.10 Dittus-Boelter equation is used to calculate:
- Radiation heat transfer
 - Conduction resistance
 - Convective heat transfer coefficient
 - Fouling factor

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 According to Kirchhoff's law, at thermal equilibrium is Absorptivity = _____
- Q.12 Write formula of Reynold number.
- Q.13 In a co-current heat exchanger, Fluids flow in _____ directions.
- Q.14 Define Conduction.
- Q.15 Write unit of heat.
- Q.16 Write one use of Heat exchanger.
- Q.17 What is reflection.
- Q.18 Write equation for convection.
- Q.19 Define insulation.
- Q.20 For a black body, emissivity is _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain natural circulation evaporators with advantages.
- Q.22 Explain Wilson plot method in heat transfer.
- Q.23 Derive the expression for Log Mean Temperature Difference (LMTD) for Parallel flow heat exchanger.
- Q.24 Derive the expression for radiant heat transfer coefficient.
- Q.25 Explain the physical significance of Nusselt number and Stanton number.
- Q.26 State and explain Newton's law of cooling.
- Q.27 Compare grey body and black body with example.
- Q.28 Derive expressions for Arithmetic mean area.